PROG7311 POE: FINAL SUBMISSION

STUDENT NAME	Nabihah Arshud Sema
STUDENT NUMBER	ST10440590
MODULE NAME	Programming 3A-Enterprise Application Development
MODULE CODE	PROG7311
LECTURER NAME	Mortda Adam
DUE DATE	8 June 2026

TABLE OF CONTENTS

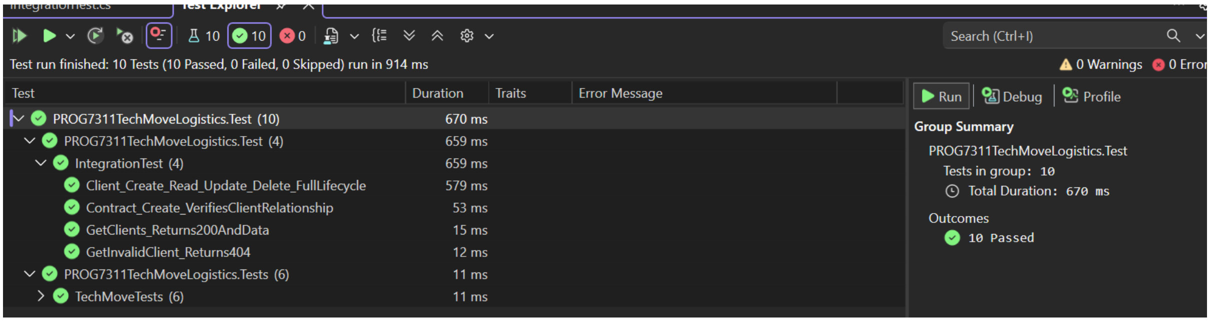
LINK TO GITHUB REPO:

https://github.com/EMKNPM/programming-3a-prog7311-part-3-Nabihah1

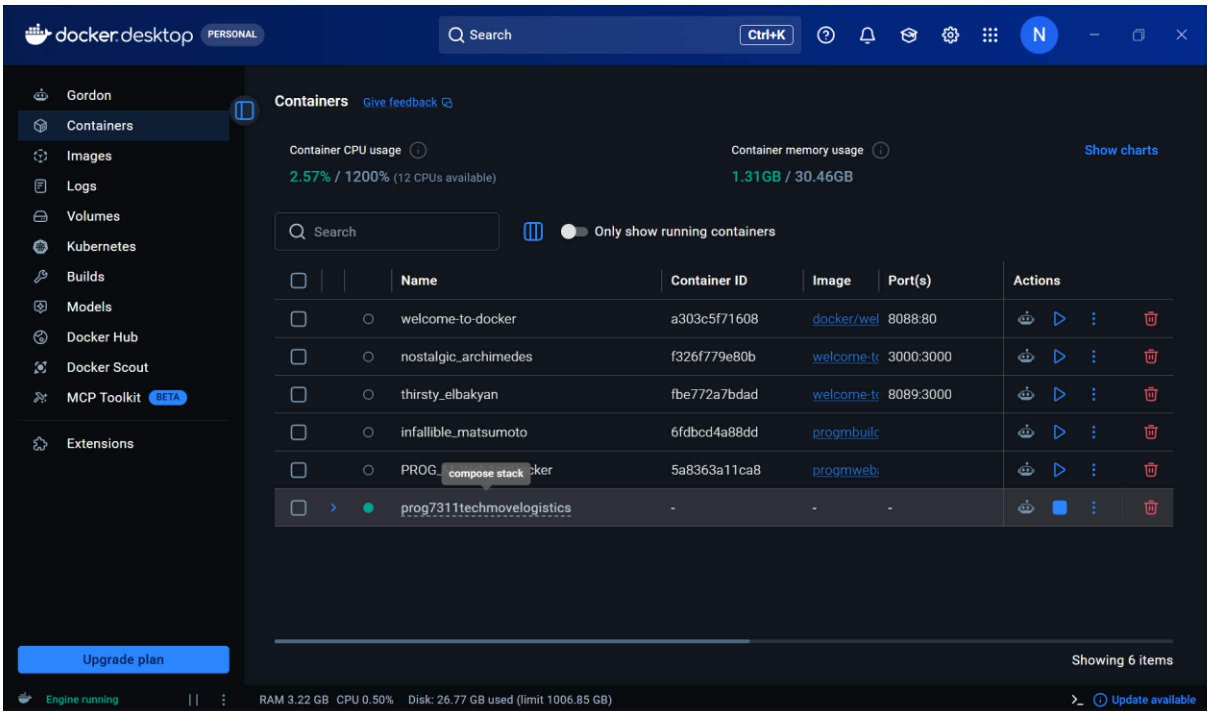
LINK TO YOUTUBE VIDEO:

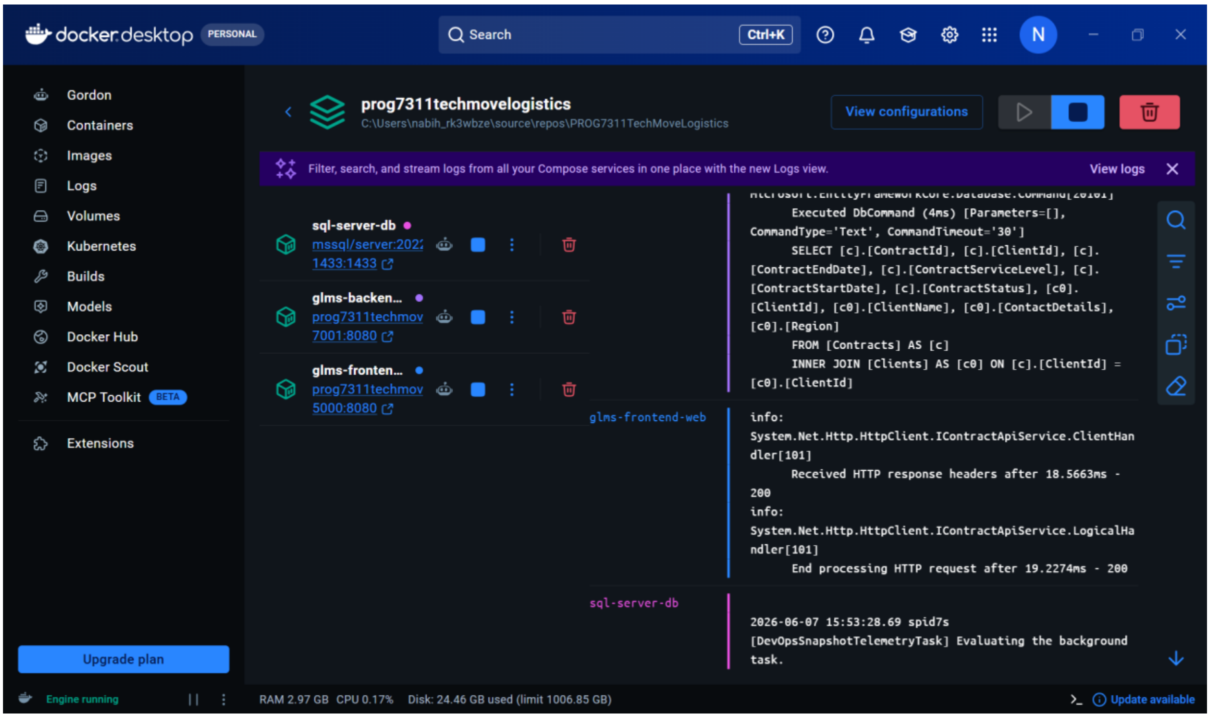
https://youtu.be/G33Cm88GzcA

SCREENSHOT OF INTEGRATION TESTS:



SCREENSHOTS OF SYSTEM IN DOCKER DESKTOP:





TECHNICAL REFLECTION REPORT

DEV OPS AND TESTING

Continuous Integration and Continuous Delivery (CI/CD) is the practise of automating software development, testing and deployment. It helps development teams to minimise human error and improve collaboration thus increasing the speed and reliability of software delivery (Inc, 2025).

Automated testing ensures that the existing functionality does not break when new changes are made to the code before software released. A good practise is to implement testing early, ideally during development after code is added or updated. This allows bugs to be detected and fixed early. Thus, preventing the, from reaching production and improving software quality from the start of the pipeline (Matick, 2025).

Automated tests check different parts of the system (e.g. Functions, APIs) to ensure everything works correctly. If any test fails, the CI/CD pipeline automatically stops deployment to prevent faulty code from being released (Inc, 2025).

Overall automated testing helps to saves time, reduce human error and ensures only tested code reaches production.

CONTAINERIZATION

Dockers packages an application together with all its dependencies (i.e. libraries, frameworks and configuration files) into a single isolated unit called a container. This ensures the application runs smoothly regardless of where its deployed (GoogleCloud, n.d).

This solves the ‘it works on my machine’ problem as differences between operating systems, software versions and system configurations are removed. Instead of relying on each machine to be set up correctly, the container carries everything needed to run the application. This ensures the application to behave the same in development, testing and production (Turner, 2026).

Developers create a Docker file that states how the environment must be built. This is used to build a Docker image which is shared and deployed across all environments. The same image is used by developers, testers and production systems thus ensuring full consistency (Aditya, 2023).

Containers are also lightweight and isolated which allows multiple applications to run on the same system without interfering with each other. This improves reliability and makes deployment more efficient and portable (Turner, 2026).

REFERENCE LIST

Aditya, 2023. *Docker ensures environment consistency, overcoming "It works on my machine" issues caused by different OS and configurations in development, testing, and production.*. [Online]
Available at: https://medium.com/@ak942389/docker-ensures-environment-consistency-overcoming-it-works-on-my-machine-issues-caused-by-e615d30cd05d
[Accessed 4 June 2026].

GoogleCloud, n.d. *What is containerization?*. [Online]
Available at: https://cloud.google.com/discover/what-is-containerization
[Accessed 4 June 2026].

Inc, Q. V., 2025 . *Test Automation for Continuous Integration/Continuous Delivery (CI/CD) Pipelines*. [Online]
Available at: https://www.linkedin.com/pulse/test-automation-continuous-integrationcontinuous-delivery-yvgxc/
[Accessed 4 June 2026].

Matick, T., 2025. *Automated Testing in CI/CD Pipelines: Bes*. [Online]
Available at: https://testmatick.com/automated-testing-in-ci-cd-pipelines-best-practices-for-reliable-software-delivery/
[Accessed 4 June 2026].

Turner, L., 2026. *What Is a Docker Container? Features and Benefits*. [Online]
Available at: https://www.theknowledgeacademy.com/blog/docker-container/
[Accessed 4 June 2026].